

FORWARD CHAINING

AIM

To write a Prolog program to implement Forward Chaining using facts and rules and derive new facts from the available knowledge.

ALGORITHM

1. Start.
2. Define initial facts.
3. Define rules for deriving new facts.
4. Check the available facts against the rules.
5. Add newly derived facts to the knowledge base.
6. Continue until no new facts can be derived.
7. Query the required conclusion.
8. Display the result.
9. Stop.

SOURCE CODE

% Initial Facts

fact(sunny).

fact(warm).

% Rules

rule(sunny, happy).

rule(warm, active).

rule(happy, smiling).

rule(active, playing).

% Forward Chaining

forward_chaining :-

 findall(F, fact(F), Facts),

 forward_chain(Facts).

```
forward_chain(Facts) :-  
    ( derive_new_fact(Facts, NewFact)  
    -> assertz(fact(NewFact)),  
        write('Derived new fact: '),  
        write(NewFact),  
        nl,  
        forward_chain([NewFact|Facts])  
    ; write('No more new facts can be derived.'),  
        nl  
    ).
```

% Find a new fact using a rule

```
derive_new_fact(Facts, NewFact) :-  
    rule(Condition, NewFact),  
    member(Condition, Facts),  
    \+ member(NewFact, Facts),  
    !.
```

% Main Program

```
:- initialization(main).
```

main :-

```
    write('--- Forward Chaining ---'), nl,  
    write('Initial facts:'), nl,
```

```
fact(F),  
write(F), nl,  
fail.
```

main :-

```
    nl,  
    write('Applying forward chaining...'), nl,  
    forward_chaining,  
    nl,  
    write('Final facts:'), nl,  
    fact(F),  
    write(F), nl,  
    fail.
```

main.

OUTPUT

```
% Prolog program for Forward Chaining  
--- Forward Chaining ---  
Initial facts:  
sunny  
warm  
  
Applying forward chaining...  
Derived new fact: happy  
Derived new fact: active  
Derived new fact: smiling  
Derived new fact: playing  
No more new facts can be derived.  
  
Final facts:  
sunny  
warm  
happy  
active  
smiling  
playing
```

RESULT

The Prolog program successfully demonstrates **Forward Chaining** by deriving new conclusions from existing facts and rules.